

CS-57 [AI / SOFTWARE ENGINEERING]

HUMAN–AI DATA ANNOTATION, PROVENANCE, AND DATASET EVALUATION PLATFORM

Project Proposal



THE UNIVERSITY OF
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Group Based 5703 Capstone Project

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ABSTRACT

The quality of an AI dataset depends not only on its final annotations, but also on whether the process used to produce those annotations is transparent and traceable. Many annotation workflows focus primarily on assigning labels while providing limited visibility into where data originated, which guidelines were applied, how human and AI contributions influenced decisions, or how disagreements were resolved. When this production history is incomplete, the quality of a released dataset becomes difficult to audit and defend.

This project aims to address this problem by extending Arc Intelligence's existing annotation platform to support a more transparent and controlled dataset-production process. The platform will combine AI assistance with human annotation and review while keeping a record of how each annotation develops from its initial suggestion or judgement to its final accepted label. It will also support the review of disagreements and check that required annotation and review information is complete before a dataset is released. In this way, AI can assist the annotation process without obscuring the human decisions that determine the final dataset.

An Agile development process will be followed throughout the project, with the development and testing of user stories carried out iteratively across each development cycle. Existing issues will be investigated and addressed alongside the implementation of new features, rather than being treated as a separate phase.

The expected outcome is an integrated annotation platform that provides greater visibility into how annotations are produced and reviewed, while supporting a structured process for AI assistance, human review, disagreement handling, and final evaluation. The core end-to-end workflow will initially be demonstrated using text annotation, with the system designed to support additional data types in future extensions.

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1. INTRODUCTION

The quality and traceability of training data are becoming increasingly important as AI systems are being used in situations where mistakes can have serious consequences. Reliable AI systems depend on high-quality datasets, but many traditional annotation platforms mainly focus on the labelling process and treat dataset production as a simple, linear process. This project aims to address this gap by building a provenance-aware data production layer on Arc Intelligence's existing platform that already handles basic annotation workflow. The system will support a full data production process with AI assistance, and the task will be managed within an organisation using role-based permissions and will automatically track each step of the production process. Our approach is to improve and extend the current system, gradually addressing its weaknesses while building the new tracking and governance features required by the project.

2. RELATED LITERATURE

2.1 Literature Review

Data annotation is an essential part of the development and assessment of Artificial Intelligence (AI) systems, as the performance of an AI system is contingent upon the data used to train and evaluate it. Additionally, data annotation is a labour-intensive task that can be expensive and challenging to coordinate when subjective tasks are involved. Consequently, research has been conducted to develop methods that integrate automation with human discretion. This review will focus on three closely related areas in the data annotation pipeline: annotation creation, annotation validation, and annotation provenance. Peer-reviewed journal and conference articles, reputable research, and technical reports published between 2008 and 2026 with more weightage given to recent contributions to AI-assisted annotation were primarily considered. Search terms included, but were not limited to, AI-assisted data annotation, human-in-the-loop machine learning, annotation quality, inter-annotator agreement, human label variation, data provenance, and dataset documentation.

2.2 AI-Assisted Annotation

Efforts to decrease the annotation effort while incorporating human discretion have been studied in the literature. Snow et al. (2008) showed that annotations created by non-experts can be beneficial when annotated quality is assessed appropriately. Active learning selects the most informative data points to be annotated by human (Settles, 2009), and weak supervision approaches such as Snorkel use indirect labelling methods to save manual annotation work (Ratner et al., 2017). More recently, human-in-the-loop machine learning has offered a framework to incorporate human discretion at any stage of the machine learning pipeline (Wu et al., 2022). The rise of large language models has also enabled AI systems to generate annotation suggestions directly. However, Felkner et al. (2024) found that using GPT-3.5 as an annotation tool had its limitations, and Schroeder et al. (2025) discovered that suggestions from LLM impacted human annotation decisions and raised annotator confidence without decreasing annotation time.

2.3 Annotation Quality and Validation

Annotation validation becomes particularly important when AI generated annotations disagree with human annotations. Nevertheless, AI is not the only source of disagreement; annotators themselves can provide different annotations for a piece of data. Plank (2022) contends that human label variation can convey valuable information, especially for more subjective annotation tasks, and that disagreement should not be treated as an error. Baledent et al. (2022) also advocate for the importance of multiple annotations and appropriate metrics when evaluating annotation quality, and Checco et al. (2017) discuss the challenges of using agreement metrics for assessing crowdsourced annotation tasks. Klie et al. (2024) additionally found that annotation quality is inadequately managed and reported in scientific publications.

2.4 Annotation Provenance and Transparency

The necessity to capture the decision-making process leads to the importance of provenance. Provenance is information that details the origin of data and its

annotation journey. For the annotation platform, provenance can capture the creator of the annotation, whether the annotation was created by a human or AI system, amendments made to the annotation, and the process of reaching the final decision. Lek et al. (2020) illustrate the benefits of capturing information about annotations and corpus development, especially when annotations are amended during the development process. This concept also applies to dataset documentation. Datasheets for Datasets proposed documenting information such as the motivation behind the dataset, how the dataset was built, characteristics of the dataset, and the intended usage of the dataset (Geburu et al., 2021). Sambasivan et al. (2021) further provide evidence that issues introduced during data collection and annotation can carry over into subsequent stages of AI system development.

2.5 Research Gap and Relevance to the Project

The literature review highlights a connected annotation workflow where AI can aid in annotation creation, validation is necessary to assess and resolve differences, and provenance is needed to capture the decision making process. While there is an abundant body of research in each of these areas, they are addressed individually rather than as part of a holistic annotation infrastructure. This project aims to fill this research gap by integrating AI annotation, human validation, cross-validation, dispute resolution, final annotation, evaluation, and provenance logging onto a single platform to enable an efficient annotation workflow and improved quality, transparency, traceability, and reproducibility of the dataset.

3. PROJECT PROBLEMS

3.1 Project Aims & Objectives

The aim of this project is to transform the existing annotation platform into a defensible dataset-production system in which every released label can be traced through its production history and verified against predefined governance requirements. The project will:

- Develop an end-to-end annotation workflow supporting AI assistance, role-based access, review, dispute resolution, and adjudication.

- Implement provenance tracking and automated release controls to support dataset integrity and reproducibility.
- Test and evaluate the system through 40–60 end-to-end scenarios, focusing on annotation quality, AI assistance, and provenance integrity.

3.2 Project Questions

1. How can an annotation platform integrate human-AI annotation while maintaining transparent decision-making and effective governance?
2. How can automated validation and evaluation ensure the quality, integrity, and reproducibility of AI-assisted datasets?

3.3 Project Scope

The project covers the annotation, governance, provenance, validation, and evaluation components of the platform. It includes project and task configuration, guideline management, dataset registration, AI-assisted annotation, human review, dispute and adjudication workflows, role-based access control, provenance recording, release validation, and dataset export.

The project will also include 40–60 end-to-end test cases covering normal and challenging scenarios. Evaluation will focus on quality, collaboration, and integrity.

The project does not aim to improve annotation speed or develop new AI models. Instead, it focuses on improving the traceability, governance, and reliability of the existing annotation process.

4. METHODOLOGIES

In this software development project, the team will extend an existing, partially built codebase, rather than building a new one from scratch.

4.1 Methods

Software Development Approach. The team follows an agile development process structured around weekly group and client meetings. Each week-long iteration will close with a demonstrable increment of the core workflow in the user interface. Within each iteration, any team member working on the existing code should follow a sequence: inspect, diagnose, validate, fix, implement, and review. To ensure consistent and traceable code management on GitHub, all code changes to the main branch should go through a pull request from an individual branch and a peer code review process.

Improving the existing codebase. At the current stage, the team has gone through a technical audit and identified 29 defects in the inherited platform. Suggested by the client, these defects should not be treated as a separate remediation phase. Therefore, our solution is attaching them to the corresponding user story whose behaviour it affects and fixing them as part of delivering that story.

AI-assisted annotation. One of the core features the team extends the existing codebase with is AI assistance, which addresses the problem that human annotations are slow and inefficient nowadays. Specifically, the system will prompt an external LLM api for a structured response. Since AI has a first pass over a task's items, users are left to review, correct and judge them with less effort. To ensure provenance, each suggestion is stored as a record of its own, attributed to the model and version that produced it, together with the annotator's response to it, including accepted, modified, rejected, or independently replaced. Overall, all history is captured at the point of decision, so that AI and human contributions remain distinguishable.

4.2 Data Collection

No new primary data will be collected by the team because all annotation items have been provided by the client. The source folder is around 4 GB with four folders, covering four key types of data available for both AI and human annotation: text,

image, audio and video. Reference annotations are provided alongside these items and are used as the ground truth.

4.3 Data Analysis

The evaluation process focuses on three dimensions specified in the project brief: *Quality*, *Collaboration*, and *Integrity*. Throughout the project, the team gradually constructs the evaluation harness and casebook. At the end of the project, the casebook should have 40 to 60 substantive end-to-end cases, including normal expected workflow, edge cases, and adversarial scenarios.

Finally, the evaluation results can be captured through this reusable casebook, designed to support regression testing across iterations and avoid reducing the measurement of dataset quality to a single unexplained score.

4.4 Deployment

The deployment process will not play a major role in our project timeline. Rather than a public release, the HEJ platform is run in a local environment for development and demonstration purposes. Specifically, the backend engine and frontend user interface are run together on the same device, consistent with the capstone being assessed through a live demo. After the project, the client will have access to the up-to-date main branch through the reviewed pull-request process described in Section 4.1 and marked with release tags.

4.5 Testing

The testing process will build on the structure already partially established in the inherited codebase, including automated backend and frontend test suites. Since the technical audit has identified tests that were failing and vacuous, those tests will be corrected and updated when new features are implemented.

Beyond unit testing, the team builds the evaluation harness and casebook described in Section 4.3 as a form of scenario testing. Each test case will be specific to the annotation workflow by exercising the full pipeline against a known expected outcome. This approach is chosen over unit testing alone because the success criterion of this project is a reliable multi-step, multi-user workflow, which unit tests cannot demonstrate on their own.

5. RESOURCES

5.1 Hardware & Software

The product is primarily a browser-based web application, with frontend accessed through modern browsers on desktop or laptop computers and backend server run on all major operating systems, including Windows, macOS, and Linux. A local GPU is not required when hosted AI services are used.

Software Technology	Project Use & Reason
Git & Github	Used for the inherited repositories, version control, issue tracking, pull requests, and code review.
Node.js, npm workspaces, Turborepo, TypeScript, Next.js, React, React DOM, Tailwind CSS, and the shared UI package	Required for hej frontend development, to implement the browser application, routing, workflow pages, and reusable interface components. Already configured in the inherited hej codebase, retaining them to avoid unnecessary build-system migration.
Python, uv, FastAPI and Uvicorn	Required for hej backend development and dependency management, REST APIs, domain services and backend tests. Already configured in the inherited hej codebase.

SQLAlchemy and SQLite	Required for hej relational data access and testing in the inherited codebase, PostgreSQL has been described as a potential production database in the handed over readme.md, but it has not yet been configured or validated in the current codebase.
OpenAI Python SDK , OpenAI-compatible endpoints, Ollama and Google Gemini API	Used for AI-assisted annotation. The inherited codebase currently calls Gemini, while labeling_ai_assistnat_mvp supports local Ollama and hosted providers including OpenAI, Qwen, Groq, DeepSeek, Doubao, and Qianfan.

5.2 Materials

Inherited source repositories and their history: The team is granted full access to hej & labeling_ai_assistnat_mvp repository, including branches, issues, setup notes, prior design decisions, and known defects, to conduct an evidence-based handover.

Client-provided curated datasets and deterministic fixtures: Realistic text, image, audio, and selected video items are needed for annotation, review, provenance, release, and evaluation demonstrations. Gold annotations are provided for accuracy, agreement, calibration and failure analysis.

5.3 Roles & Responsibilities

Shared responsibilities

Throughout the semester, all team members will produce an integrated and demonstrable deliverable, maintain tests, documentation, and change notes for their work, keep a decision record for significant design and technical choices, including the selected option and alternatives considered, participate in pull-request code review, contribute to the shared 40-60 case evaluation casebook, and keep the main integration branch up to date and runnable.

Individual roles and corresponding responsibilities

- **Project Leader & Integration Coordinator (Hanchen, Yi):** Manage scope, milestones, risks, client communication, meetings, cross-repository coordination, and the runnable demonstration.
- **HEJ Frontend Developer / UI Tester (Dishank, Yi, Kanishka):** Own hej/apps/hej-web; fix interface defects and maintain task, annotation, review, history, release, state-handling, and component-test code.
- **HEJ Backend / Database and Provenance Developer (Tim, Yi, Parth):** Own hej/apps/hej-api; maintain APIs, persistence, authentication, workflow transitions, provenance, and release checks.
- **AI Annotator / HEJ AI Integration Developer (Hanchen, Michael, Jingwei):** Maintain labeling_ai_assistnat_mvp analyzers, model selection, structured multimodal outputs, and candidate integration with provider, model, and error metadata.
- **Evaluation Developer / Data Analyst (Tim, Michael, Parth):** Define evaluation metrics and analysis, consolidate the casebook contributed by all members, and evaluate quality, AI influence, provenance completeness, and release reproducibility.

6. EXPECTED OUTCOMES

6.1 Project Deliverables

This project's deliverables are committed at the level of epics rather than individual user stories. The team maintains a backlog of 59 stories grouped into eleven epics, and that backlog is a working planning instrument rather than a fixed specification: stories will be split, added and re-sequenced as the inherited codebase becomes better understood. Committing the proposal to a fixed story list would either bind the team to a plan already known to be provisional or allow scope to drift silently as the stories beneath it change. Committing at epic level fixes what the

project will deliver and how completion will be judged, while leaving the team free to adapt how each epic is decomposed. Each deliverable therefore carries a definition of done assessable independently of the stories that produce it, and the deliverables are sequenced so that each depends only on those above it.

D1: Engineering foundations and working practice (Epic A, W5–6)

D2: Core end-to-end annotation workflow for text data (Epics B–E, W7–10)

D3: Provenance and governance layer (Epics F–G, W9–10)

D4: Release validation and export (Epic H, W10–11)

D5: Evaluation harness and casebook, 40–60 cases (Epic I, W10–11)

D6: Documentation and handover evidence (Epic K, W12)

D7: Assessed academic artefacts (W5, W9, W13)

Definitions of done, epic mapping and week-by-week detail for each deliverable are set out in [Appendix \[6\]](#).

6.2 Implications

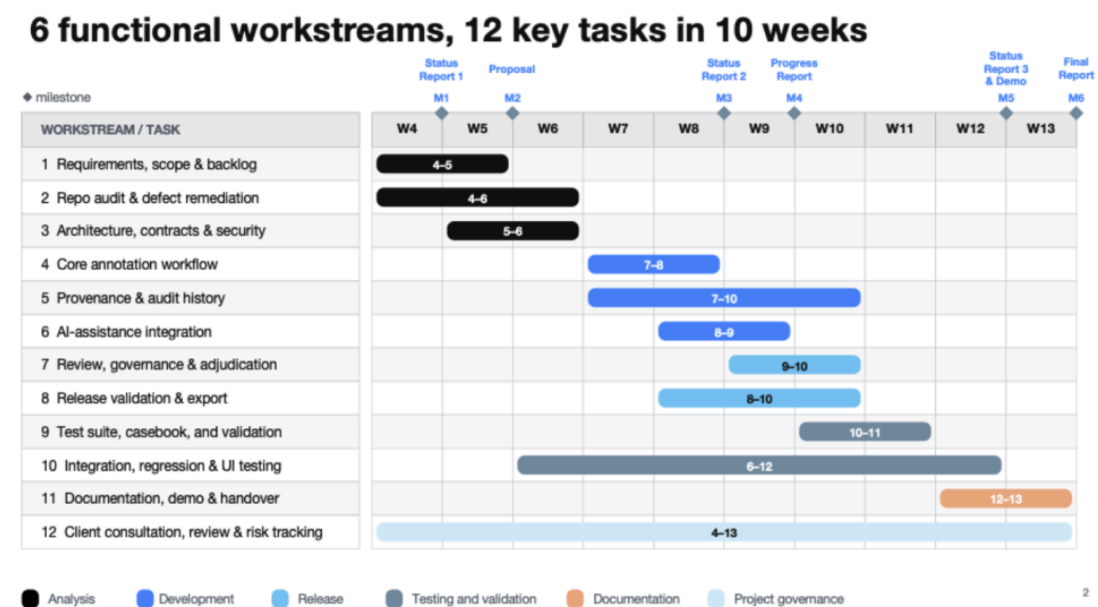
For the client. The team's technical audit found the platform is not currently safe for a real multi-user pilot: finalised annotations can be silently overwritten, users can edit each other's work, annotators can approve their own submissions, and a single item can export more than one conflicting answer. Completing these deliverables moves the platform from a demonstrator to a system whose outputs can be defended to a third party. The evaluation harness and defect register also outlast the project, lowering the cost to whoever takes the codebase on next.

For the domain and a wider audience. Human-in-the-loop annotation, AI-assisted labelling, quality management and provenance have each been studied substantially but largely in isolation. This project contributes a working implementation in which they operate as one connected pipeline - and, more broadly,

demonstrates one way that accountability for how a dataset was produced can be enforced by a system rather than asserted by its operators.

Limits. The workflow is demonstrated against text data on a local instance at capstone scale. No models are trained or fine-tuned, and findings on AI influence on human judgement derive from the project's own casebook rather than a controlled study.

7. MILESTONES / SCHEDULE



The Gantt chart above presents the planned work across six interconnected workstreams as indicated in the legends section. They are organised into weekly cycles aligning to the project management methodologies and reflect the continuous nature of our project. Together, these workstreams helps understand and stabilise the inherited codebase, implement the proposed functionality, validate the integrated workflow, and deliver a demonstrable system within the 10-week timeframe.

A detailed plan of activities and milestones have been attached in the [Appendix \[7\]](#).

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9.

APPENDIX

Section 6: Expected Outcomes - Deliverables Breakdown

#	Deliverable	Definition of done	Epics	Target
D1	Engineering foundations and working practice - a reproducible development environment, trustworthy test suites, credential hygiene, and a pull-request and code-review workflow	Any team member can build and run the platform from a clean clone; frontend typecheck and both test suites pass and test what they claim to; the committed credential is rotated and removed as a source default; every change to main arrives through a reviewed pull request	A	W5-W6
D2	Core end-to-end annotation workflow for text data - project creation, policy configuration, task setup, dataset ingestion, AI-assisted first pass, human review, and dispute resolution through to final judgement	A text annotation item can be carried from project creation to adjudicated final judgement through the user interface, without manual database intervention	B, C, D, E	W7-W10
D3	Provenance and governance layer - full production history per label, and a capability-based permission model	Every accepted label records the guideline version in force, annotator and reviewer identities, the AI suggestion where	F, G	W9-W10

		applicable, the human response to it, and the final adjudicated judgement; annotators, reviewers, arbitrators and task owners hold distinct capabilities, and the same user cannot annotate and review the same item		
D4	Release validation and export - an automated gate preventing release of incomplete or untraceable datasets	Export is refused where items are unreviewed, provenance is incomplete, or versions are incompatible; a permitted release produces exactly one authoritative answer per item together with a manifest sufficient to reproduce it	H	W10-W11
D5	Evaluation harness and casebook - a reusable suite of 40–60 end-to-end cases including adversarial scenarios	The casebook runs repeatably against the platform and reports on quality, collaboration and integrity; adversarial cases covering confidently wrong AI suggestions, ambiguous items, annotator disagreement and post-annotation version change are included	I	W10-W11

D6	Documentation and handover evidence - technical documentation, per-member decision records, the maintained defect register, and a statement of known limitations	A reader outside the team can understand, run and extend the system from the documentation alone; each member can evidence and defend their own design decisions; unresolved issues and failed approaches are recorded as deliberately as successful ones	K	W12
D7	Assessed academic artefacts - proposal report, progress report, final report and technical report, final presentation and live demonstration	Each artefact is submitted by its published deadline; the final demonstration executes the complete workflow against the client-provided dataset	N/A	W5, W9, W13

Section 7: Milestones / Schedule - Detailed Activity Schedule

The project is organised into weekly cycles of analysis, implementation, review, and validation.

Six key milestones provide measurable checkpoints for proposal and progress reporting, client review, workflow demonstrations, system validation, and final delivery. Each milestone is linked to defined outputs so value creation is clear. If a milestone is delayed, lower-priority interface enhancements will be deferred before essential requirements are reduced.

Regular communication with Arc Intelligence will continue throughout the project, with more frequent and ad hoc consultations during development, integration, and testing. These consultations will confirm that implemented functionality remains aligned with client expectations and will support early decisions where technical limitations or scope trade-offs arise. A detailed milestone and task table following

the gantt chart identifies the activities, expected outputs, and planned timing required for on-time project completion.

Milestone	Tasks	Reporting	Date
Week-1	- Project kick-off - Team introductions	None	-
Week-2	Basic research on the project topic based on description given	Client meeting to review the work plan	-
Week-3	- Repository access setup - Workspace setup	None	
Week-4	Milestone: Status report 1 - Kick-off meeting and scope agreement with client - Repository and codebase review - independent technical audit (29-defect list and fix plan) completed	Vision and expectation workshop with client	
Week-5	Milestone: Proposal Report - Translate agreed scope into user stories - Establish and communicate pull-request and code-review workflow - Conduct schema/contract audit		
Week-6	- Validate and correct contracts between front-end, back-end, and AI service (where applicable) - Validate and amend schema where needed - Fix cross-project write-scoping defects	Progress update with client	

Week-7	● Implement core annotation backend logic including: ○ Canonical-annotation model ○ Atomic draft ○ Task lifecycle management	None	
Week-8	● Milestone: Status report 2 ● Implement core annotation backend logic including: ○ Dataset ingestion transactionality ○ AI module integration ○ Export/completion rule alignment	Progress update with client	
Week-9	● Milestone: Progress Report ● Implementation of features ○ Human review workflow ○ Role-based governance model ○ Role related features and logics for project governance workflow (annotator/reviewer/expert/admin permissions, prohibition of self-review, dual sign-off / expert-gate / arbitration-ready policies)	Progress update with client	
Week-10	● Implementation of features ○ Dispute resolution process and escalation workflow (cross-validation disagreement routing, expert adjudication, dispute route-back)	Client meeting to review the logics implemented	

	• Development of evaluation harness framework		
Week-11	• Demo of end-to-end integration with provided datasets • Finalise the complete evaluation casebook with 40-60 cases, including adversarial cases • Conduct regression testing	Progress update with client	
Week-12	• Milestone: Status report 3 • Milestone: Demo • Security review and hardening • Bug reviews • Documentation • Technical report drafting (per-member decision record) • Final demo preparation work	Client meeting to showcase the demo	
Week-13	• Milestone: Final presentation • Final report submission • Technical report submission	Client meeting for project presentation and walkthrough	